

Total No. of Questions : 8]

[Total No. of Printed Pages : 1

Roll No

EC-402 (GS)

B.E. IV Semester Examination, June 2020

Grading System (GS)

Electromagnetic Theory

Time : Three Hours

Maximum Marks : 70

Note : i) Attempt any five questions.

ii) All questions carry equal marks.

- Point charges $1\text{ }\mu\text{C}$ and $2\text{ }\mu\text{C}$ are located at $(3, 2, -1)$ and $(-1, -1, 4)$ respectively. Calculate the electric force on a $10\text{ }\mu\text{C}$ charge located at $(0, 3, 1)$ and the electric field intensity at that point.
- Verify the divergence theorem, for an electric field intensity $\vec{E} = \frac{\rho r}{3\epsilon} \vec{a}_r$ for $r \leq a$ in spherical coordinate. For spherical surface choose $r = c \leq a$.
- Two parallel conductors of infinite extent are carrying currents of 10A and 20A in opposite directions. If the separation between the conductor is 10cm , calculate the force per unit length experienced by either conductor.
- Derive Maxwell's equations in:
 - Integral form
 - Differential form
 - Free space
 - Harmonically time-varying fields
- Find the polarization of a wave if the electric field intensity is given by
$$\vec{E} = (-j25\vec{a}_x + 25\vec{a}_z)e^{-(0.01 + j120y)} \frac{V}{m}$$
- Explain what is skin depth? Calculate α , β , γ , λ , Y_p and n for a damp soil at $f = 10^6\text{ Hz}$. It is given that for damp soil $\epsilon_r = 12$, $\sigma = 2 \times 10^{-2}\text{ mhos/m}$, $\mu_r = 1$. where η = intrinsic impedance, λ = wavelength.

OR

 - Derive wave equations for non-conducting medium?
 - Calculate the phase velocity and the magnitude of the attenuation constant of plane wave at a frequency of 10 GHz in polyethylene. It is given that $\mu = \mu_0$, $\epsilon_r = 2.3$ and $\sigma = 2.56 \times 10^{-4}\text{ mhos/m}$.
- Calculate the magnetic flux density produced by an infinite thin long wire carrying current I at distance R .
- Answer any four of the following:
 - What does the divergence of a vector signify?
 - State the Biot-Savart law in your own words.
 - Write Maxwell's equations in point form.
 - What is the major difference between an elliptically and circularly polarized wave?
 - What happens if the angle of incidence is greater than the critical angle?
 - Explain the difference between phase velocity and group velocity.
